

Water Vendo wiring schematic

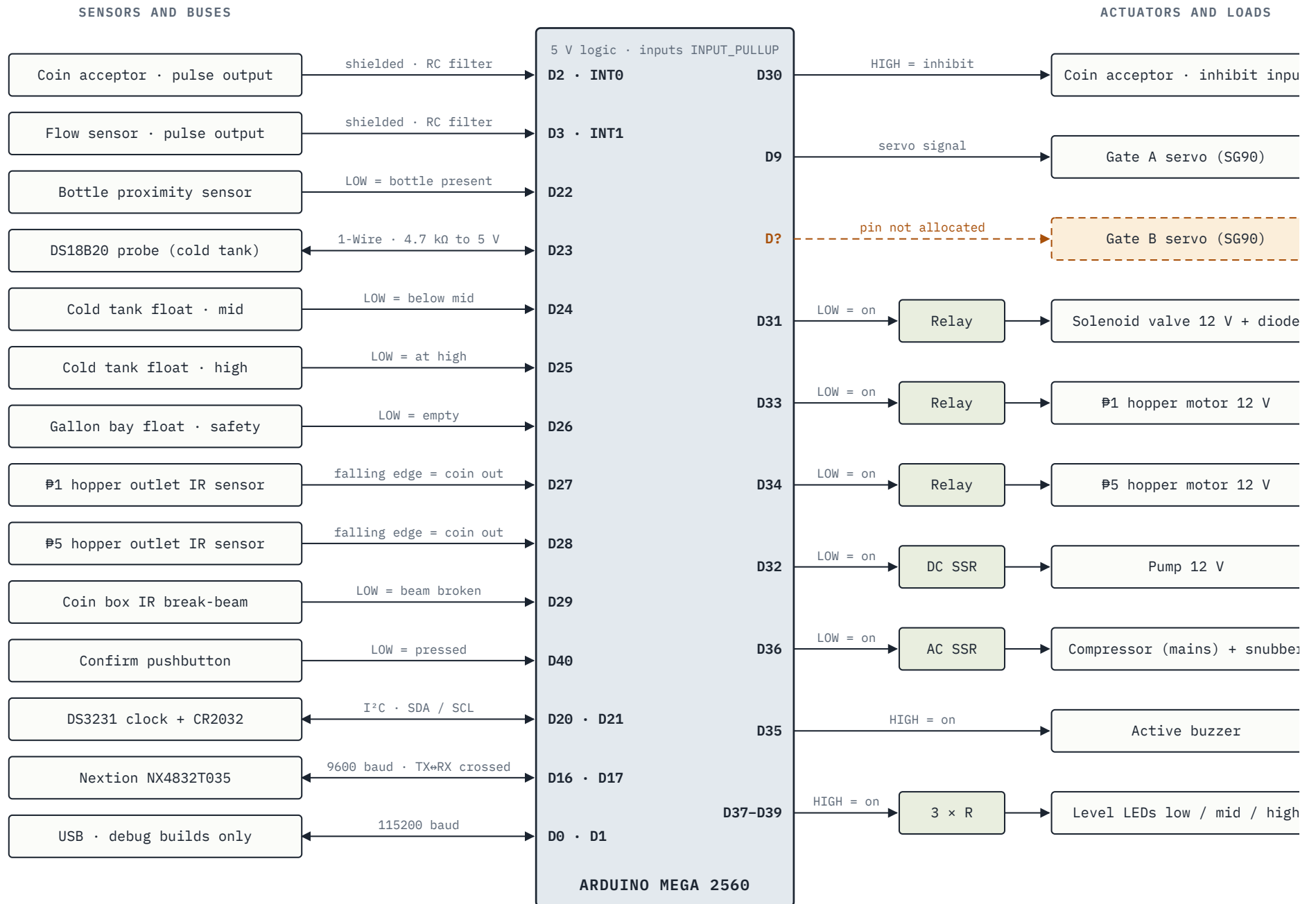
Interconnect wiring for the coin-operated water refill machine: what connects to which pin of the controller, how each load is switched, and how power and ground are distributed.

PROJECT EMX-2026-WATERVENDO-01	PREPARED BY eMathrix Technologies	CONTROLLER Arduino Mega 2560, 5 V	SOURCE docs/wiring.md, include/config.h, Fab Drawing Rev B	DATE 2026-10-04
STATUS Draft for review. Not built, not bench-verified				

Open items to settle before this is used for the build

- Gate B servo has no pin yet.** Fabrication drawing Rev B replaces the single three-position diverter servo with two SG90 gate servos. The firmware pin map still lists one servo on D9. This sheet shows Gate A on D9 and Gate B as unallocated (dashed). The pin must be assigned by the Rev B work order and added to config.h and wiring.md together.
- Coin acceptor supply voltage.** wiring.md puts the acceptor on the 5 V rail. Rev B names a CH-926 class acceptor, and that class commonly runs on 12 V. Confirm against the unit actually bought before wiring it.
- Coin box beam height.** wiring.md says the beam sits at the fill line. Decision D-8 moved it to about 80% of the usable depth. The wiring is the same; the mounting position differs.
- Not specified anywhere yet:** supply ratings, fuse values, wire gauges, connector types, LED resistor values, and which relay-board channel carries which load.
- Mains wiring** (supply inlet, compressor leg, AC SSR) must be done and checked by a qualified electrician.

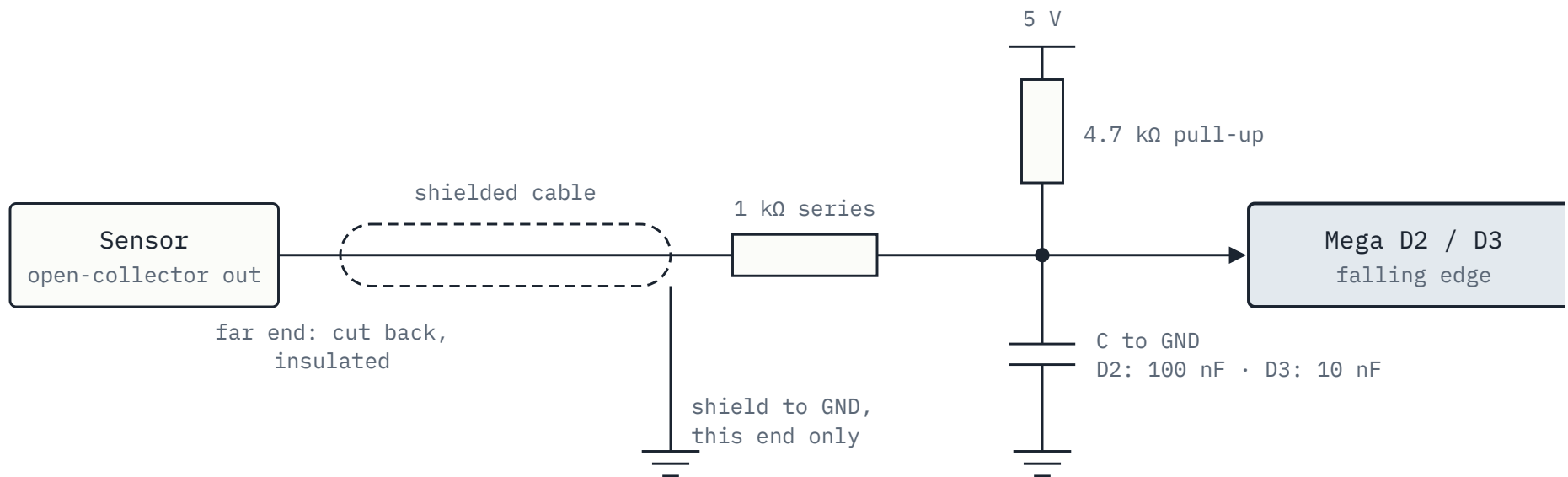
1. Controller connections



Sensors and data buses on the left, loads on the right. Every switch-type input is wired to ground and reads LOW when active, so a broken wire reads as inactive. The 12 V and mains loads are never driven from a Mega pin directly. The dashed item is the Rev B Gate B servo, which has no pin assigned.

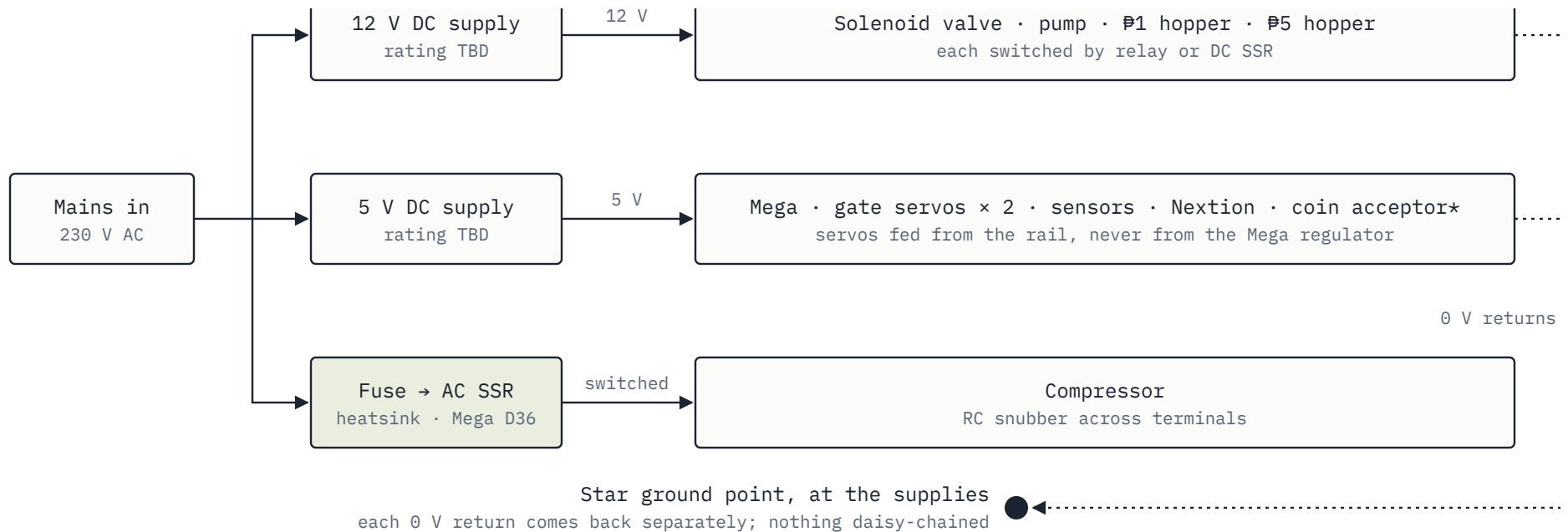
2. Pulse input filter (D2 and D3)

These are the only two lines where one false edge is a money or volume error, so both get the same protection.



Coin pulse D2 uses 100 nF (corner about 1.6 kHz). Flow pulse D3 uses 10 nF (about 16 kHz). Do not increase either capacitor, and do not raise the series resistor above about 1.5 kΩ.

3. Power and grounding



* coin acceptor supply voltage to be confirmed against the purchased unit

Two DC rails plus a separately fused mains leg for the compressor. The dotted lines are the 0 V returns, brought back to a single star point so motor current does not share a path with the coin pulse ground.

4. Pin assignment

PIN	SIGNAL	DIRECTION	ACTIVE LEVEL	INTERFACE
D0 / D1	USB serial, debug builds only	in/out	115200 baud	—
D2 (INT0)	Coin acceptor pulse	in	falling edge	shielded, RC filter, 4.7 kΩ pull-up
D3 (INT1)	Flow sensor pulse	in	falling edge	shielded, RC filter, 4.7 kΩ pull-up
D9	Gate A servo signal (was: single diverter servo)	out	PWM	servo power from 5 V rail
not allocated	Gate B servo signal (Rev B)	out	PWM	pin to be assigned by work order
D16 (TX2)	Nextion RX	out	9600 baud	common ground
D17 (RX2)	Nextion TX	in	9600 baud	common ground

D20 / D21	DS3231 SDA / SCL	in/out	I ² C	module pull-ups only
D22	Bottle proximity	in	LOW = present	INPUT_PULLUP
D23	DS18B20 temperature	in/out	1-Wire	4.7 kΩ to 5 V
D24	Cold tank float, mid	in	LOW = below mid	INPUT_PULLUP
D25	Cold tank float, high	in	LOW = at high	INPUT_PULLUP
D26	Gallon bay float (safety lockout)	in	LOW = empty	INPUT_PULLUP
D27	₱1 hopper outlet count	in	falling edge	IR slot sensor
D28	₱5 hopper outlet count	in	falling edge	IR slot sensor
D29	Coin box full	in	LOW = beam broken	IR break-beam
D30	Coin acceptor inhibit	out	HIGH = inhibited	direct
D31	Solenoid valve	out	LOW = on	relay, flyback diode at valve
D32	Pump	out	LOW = on	DC SSR or MOSFET module
D33	₱1 hopper motor	out	LOW = on	relay
D34	₱5 hopper motor	out	LOW = on	relay
D35	Buzzer	out	HIGH = on	active buzzer
D36	Compressor	out	LOW = on	AC SSR, zero-cross, heatsink, snubber
D37 / D38 / D39	Level LEDs low / mid / high	out	HIGH = on	series resistor each
D40	Confirm button	in	LOW = pressed	INPUT_PULLUP
A0–A15	Unused	—	—	spare

5. Build notes

- Fit a flyback diode directly across the solenoid valve.

For the AC compressor, use a DC SSR or MOSFET module with a heatsink and snubber.

- The pump needs a DC SSR or MOSFET module. An AC SSR will not switch DC.
- Size the compressor SSR for at least three times the running current, mount it on a heatsink, and fit an RC snubber across the compressor.
- Power both gate servos from the 5 V rail, not from the Mega's regulator. Gate A must rest on the hopper path with power off (Rev B, sheet 2).
- Use a DS3231 module, not a DS1307 or DS3231M, with a good CR2032. If the module has a charging circuit, remove its resistor or fit an LIR2032.
- Use the waterproof stainless DS18B20 probe.
- Nextion TX goes to Mega RX2 (D17) and Nextion RX to Mega TX2 (D16), with a common ground.
- Nothing uses D44–D46: the Servo library takes Timer5.